

Erdős Problem #956

EP-956

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

If $C, D \subseteq \mathbb{R}^2$ then the distance between C and D is defined by $\delta(C, D) = \inf_{\substack{c \in C \\ d \in D}} \|c - d\|$. Let $h(n)$ be the maximal number of unit distances between disjoint convex translates. That is, the maximal m such that there is a compact convex set $C \subset \mathbb{R}^2$ and a set X of size n such that all $(C + x)_{x \in X}$ are disjoint and there are m pairs $x_1, x_2 \in X$ such that $\delta(C + x_1, C + x_2) = 1$. Determine $h(n)$ - in particular, prove that there exists a constant $c > 0$ such that $h(n) > n^{1+c}$ for all large n .

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>